

California State University, Northridge
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Senior Design Final Project Report

FM Transmitter

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<u>Abstract</u>	
I. <u>Introduction</u>	
II. <u>Project Description</u>	
Specifications.....	
Block diagram.....	
Description of Subassemblies.....	
Complete Schematic.....	
III. <u>Design Methodology</u>	
Standards.....	
FCC Regulations.....	
Cost Analysis.....	
Schedule.....	
IV. <u>Test Plan</u>	
Results.....	
Conclusion.....	
V. <u>Appendix</u>	
Parts List.....	
Equipment and Software.....	
References.....	
Calculations.....	

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Abstract – The goal of this project is to develop a working FM transmitter that meets certain design specifications and functions well as a final, built product. Frequency modulation of a signal encodes information in a carrier wave by varying the instantaneous frequency of the wave. This project culminates in the successful construction of a working FM transmitter made from circuit components such as inductors, capacitors, resistors and transistors. The transmitter consists of an audio input in the form of an AUX cable, an oscillator circuit to generate the radio frequency signal, an amplifier stage to amplify the signal, and a radio frequency stage to modulate the signal to the correct FM band, and a varactor diode to match the impedance of the transmitter to the impedance of the antenna to efficiently transfer power and prevent the circuit from overheating. The finished transmitter's success demonstrates the students' proficiency in the subject matter which covers both circuit design and frequency modulation. The design of this transmitter within specific working characteristics mimics real-world requirements of electrical engineers where designs have certain goals and restrictions for the purpose they are intended for. Simulation, calculation, and real-world testing were all essential to the success of this project. Common design choices were made for a number of components in the design, but for others, calculation and simulation was essential in the

selection of components to create a working final product.

Index terms – FM Transmitter, Oscillator, Pre-Amplifier, Antenna, FM Band, Frequency Modulation, Carrier Wave, Shunt-Shunt Feedback, Current-Voltage Feedback.

I. Introduction:

An FM transmitter is a device that generates an alternating current on a radio frequency (FM band covers 88 to 108 MHz) to produce frequency waves, usually to modulate an audio signal so that it can be transmitted on an FM band and picked up by a tuned receiver.

The first radio transmitter was created by Heinrich Hertz in 1887 (called Hertzian oscillators). The technology was later improved to transmit Morse code by Guglielmo Marconi in 1895. In the 1920s, vacuum tube transmitters were created, using feedback oscillators to produce continuous waves that could be modulated to recreate sound waves on the AM band. FM transmission, later invented by Edwin Armstrong in 1933, was then the preferred method for transmitting audio due to its resistance to noise and interference. The invention of radar in World War II placed a higher focus on the development of this technology, and the later invention of the transistor in the 1960's allowed the technology to scale down into a more portable form, which is the type of transmitter this project focuses on.

II. Project Description:

Specifications:

1. Transmits FM signal at least 100 feet
2. Transmits CD player output to FM tuner
3. Receiving range of 88-108 MHz
4. A physical constraint of less than 4 inches on any side
5. Powered by either a 12 V car battery, a 12 DC supply and/or have its own power capability.
6. Conform to all FCC regulations
7. Its max interference to other station signals of -90 dB down

Block Diagram: Before choosing a final design, it was necessary to develop a basic block diagram of the sub-assemblies needed to build an FM Transmitter.

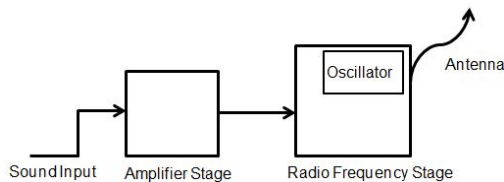


Figure 2: Block diagram showing the components of an FM Transmitter.

Description of Subassemblies: A common emitter amplifier was used as the pre-amplifier stage. This causes an increase in the amplitude of the incoming signal. A common emitter amplifier was chosen because it has high gain, bandwidth, high input resistance and low output resistance.

The next stage is the Radio Frequency stage, which contains a second transistor and an LC oscillator.

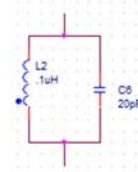
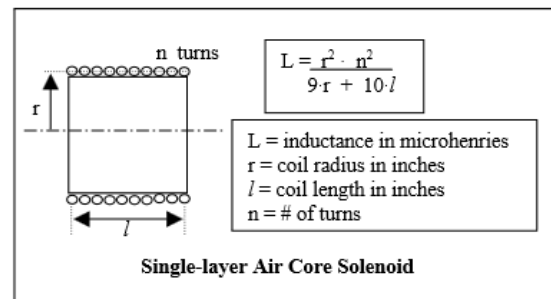


Figure 7: LC oscillator circuit.

In LC circuit, the inductor was made from a piece of copper wire using the Wheeler formula:

Air-Core Inductor Formulas (Wheeler Approximations)



The inductor is a 6-turn coil that was made by winding an 18 gauge copper wire around a 1/4 inch thick bolt with length of 0.5 inches. **Inductance is equal to 0.09184 μ H.**

$$f_{osc} = \frac{1}{2\pi\sqrt{LC}}$$

The **trimmer capacitor** can be adjusted from **6 to 40pF**. With a capacitance (C) of 6pF, the frequency is filtered to 214.4 MHz, and using a C equal to 40 pF, the frequency becomes 83 MHz. Using the trimmer capacitor, the transmitter is capable of fully covering the FM spectrum of 83 MHz through 108 MHz.

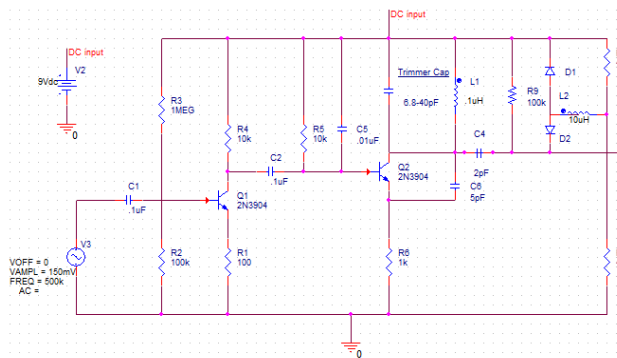
By selecting a C of 15pF and an L of 0.1 μ H, the oscillation frequency is 97.8 MHz

The third stage of our design involves a varactor diode, which matches the impedance of the transmitter to the impedance of the antenna. This allows the transfer of power to the antenna to be as efficient as possible, and prevents the

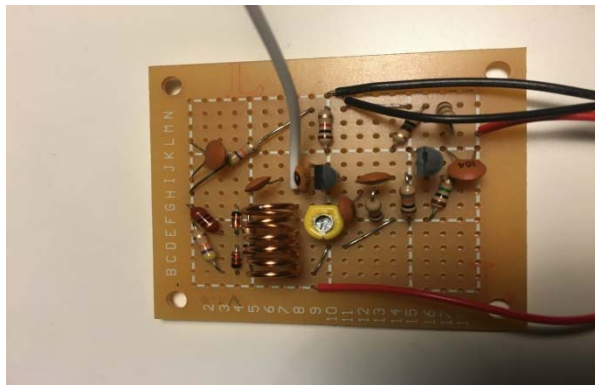
circuit from wasting power and overheating due to the standing waves. The final stage of the FM Transmitter is the antenna which broadcasts the modulated frequency into the air as electromagnetic waves.

Complete Schematic:

The following figure shows the complete circuit schematic implemented in PSpice.



Complete schematic of FMT



Circuit Design of FMT

III. Design Methodology: The primary goals in the design process were to meet the specifications and to build an efficient circuit without over-designing. The completed circuit was to be simple and reliable. The number of components was kept to a minimum so that the corresponding size would be small, resulting in a lower cost as well as make troubleshooting individual components easier.

This project involved research, cost analysis, and creation of a design schedule. Research and hand calculations were used to determine if it was even possible to meet the specifications. These hand calculations were used to design the FM Transmitter, and PSpice was used for simulation. The circuit was then soldered and tested. Breadboards were not used for this design, since the built-in breadboard capacitance would have interfered with the expected results.

Standards: In order to meet national engineering standards only EIA (Electronic Industries Association) parts were used in this design. To comply with this standard, only standard value resistors with $\pm 5\%$ error and standard value capacitors with $\pm 20\%$ error were used. This means that common parts from varying manufacturers will be interchangeable within this discrete FM Transmitter design.

FCC Regulations: In the US, The FCC specifies that no license is needed if the FM transmitter has a maximum coverage radius of 200 ft, which is well within this project's scope. However, the maximum effective radiated power has a limit of 0.01 microwatts, or more specifically 250 uV/meter measured at 3 meters.¹

Cost Analysis:

(6) resistors at \$0.01 each	\$0.06
(6) capacitors at \$0.05 each	\$0.30
(1) adjustable capacitor	\$0.40
(2) transistors at \$0.50 each	\$1.00
(1) input plug	\$0.50

¹ ["Permitted Forms of Low Power Broadcast Operation"](#) (PDF).

(1) solder board \$0.50
 (1) 9V battery clip.....\$0.50
 solder and wire \$0.25
 Total: \$ cost per unit

IV. Test Plan: To test the complete design, sound was input to the mono phone plug and the radio waves were transmitted to a radio receiver.

The transmitter was tested to make sure it covered the complete FM range of 88MHz-108MHz and maximum transmitting distance was measured.

Results: The results obtained are displayed in the table below.

receiving range	88-108 MHz
max transmitting range	150 feet
input	mono phone plug
Size	4"x2"x1"
power supply	9 volts
max interference to other stations	-90 dB down
Max Emission Limit	-61.50 dB

Table 1: Summary of results.

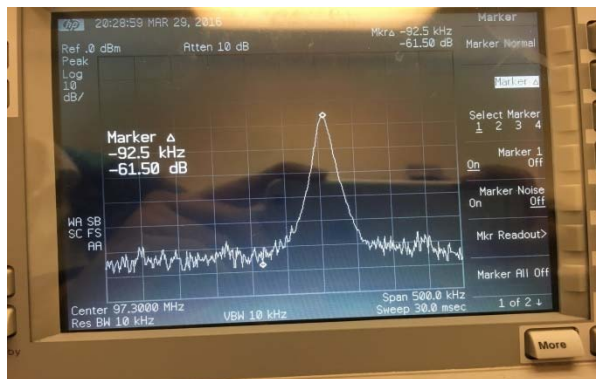


Figure 12: Maximum Signal Emissions on Spectrum Analyzer.

Conclusion:

The results depicted above show that the FM transmitter matches the specifications. The receiving range for the output signal was around 88 – 108 Mhz. Furthermore, it is less than 4 inches on any side, has a self-contained power supply less than 12Vdc, has a max interference to other stations' signals of -90 down and it complies with FCC regulations by having a maximum emissions limit of -61.50 dB.

V. Appendix:

Parts list:

QTY	Component	Value	Type
1	resistor	100Ω	Carbon, +/-5%
1	resistor	1kΩ	Carbon, +/-5%
2	resistor	10kΩ	Carbon, +/-5%
2	Resistor	47kΩ	Carbon, +/-5%
2	Resistor	100kΩ	Carbon, +/-5%
1	capacitor	2pF	
1	capacitor	5pF	
3	capacitor	0.1μF	
1	capacitor	0.01μF	
1	capacitor	6.8-40pF	Trimmer
2	Transistors	2N3904	
2	diode		
5"	copper wire	18 gauge	
8"	hook-up wire	22 gauge	
1	battery	9V	
1	Mono-phone-plug	1/8"	

Table 2: Parts used in the FM Transmitter.

Equipment and Software:

Type	Serial No.
Spectrum Analyzer	US39032586
Oscilloscope	CN50534339
Power Supply	81208767
PSPICE	Version 16.0.0.s001

Table 3: Equipment and software used.

References:

^ B. Boashash, editor, "Time-Frequency Signal Analysis and Processing – A Comprehensive Reference", Elsevier Science, Oxford, 2003; ISBN 0-08-044335-4

B. P. Lathi, *Communication Systems*, John Wiley and Sons, 1968 ISBN 0-471-51832-8, p, 214–217

"Permitted Forms of Low Power Broadcast Operation" (PDF). https://apps.fcc.gov/edocs_public/attachmatch/DOC-297510A1.pdf. Federal Communications Commission. *Retrieved 26 March 2016*.

Calculations:

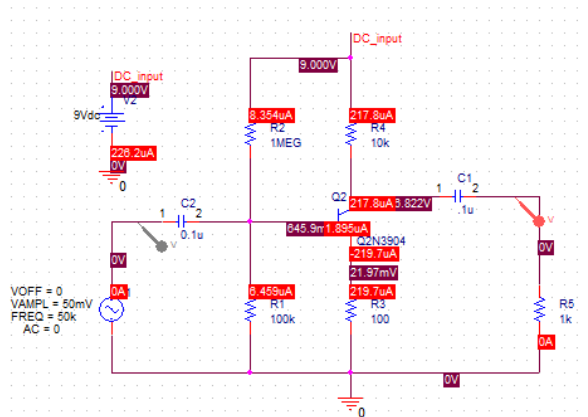
Amplifier Calculations (Stage 1):

Complete FM Transmitter

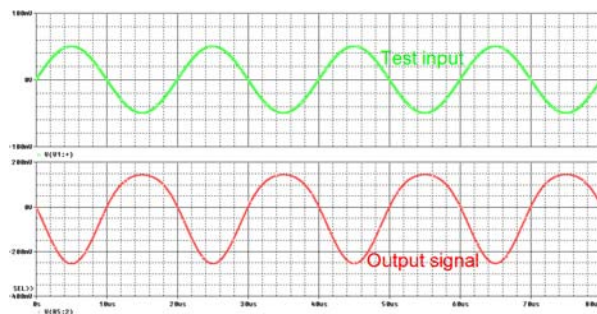
Complete FM Transmitter with currents and voltages

PreAmp Gain= for PreAmp stage with complete circuit connected

Pre-Amp stage:



pre-Amp stage with currents and voltages



$$\text{PreAmp Gain} = \frac{V_{out}}{V_{in}} = \frac{375\text{mV}_{p-p}}{100\text{mV}_{p-p}} = 3.75$$

With $R_L = 5.2 \text{ k}\Omega$ in place of the rest of the circuit. $\Rightarrow$ **Beta** = 200.

DC Analysis:

Gain Calculation:

$$A_v = \frac{g_m R_L'}{1 + g_m R_{EAC}} = \left(\frac{I_{C1}}{V_T} \right) (R_C // R_L) / \left[1 + \left(\frac{I_{C1}}{V_T} \right) (R_{EAC}) \right]$$

$R_L = R_{in}$ for 2nd transistor

$$R_L = R_{\pi} = R_L = r_{\pi} = \frac{\beta}{g_m}$$

$$r_{\pi} = \frac{200}{I_{C2}/V_T} = \frac{200}{1\text{mA}/26\text{mV}} = \frac{200}{0.0384} = 5.21 \text{ k}\Omega$$

$A_v =$

$$\left(\frac{1\text{mA}}{26\text{mV}} \right) (10000\Omega // 5.200\Omega) / \left[1 + \left(\frac{1\text{mA}}{26\text{mV}} \right) (100) \right]$$

$$= (0.0385)(3421\Omega) / (4.85) = 27.2\text{V/V.}$$

However, PSPICE shows **$A_v = 1.75\text{V/V}$** .
Solution: stage 2 is shunt-shunt so use this formula:

$$R_L = R_{in2} = R_{ic} = \frac{R_{io}}{1 + \beta A_o} = 30\Omega$$

Therefore, **$A_v = 3.75\text{V/V}$**

Oscillator Calculations:

$$f_{osc} = \frac{1}{2\pi[\text{sqrt}(LC)]}$$

$$= \frac{1}{2\pi[\text{sqrt}(0.1\mu)(15p)]} = 97.8 \text{ MHz}$$

$$L = \frac{(R^2)(T^2)}{9R + 10l}$$

L =inductance in μH

R =radius of 1 coil in inches

T = number of turns

l =length coiled inductor in inches

at the frequency = $1/2\pi \cdot \text{sqrt}(LC)$, in terms of all the components in the oscillator stage:

$$\text{Given: } R_{eq} = r_{\pi} // R_6$$

$$C_{eq} = C_{out} + C_5$$

$$Z_{eq} = sL // \frac{1}{sC_{eq}} = \frac{sL}{1 + s^2 LC_{eq}}$$

Node 1:

$$\frac{-V_{BE}}{R_{eq}} - V_{BE}(sC_{in}) + (-V_{BE} - V_C)sC_6 = gmV_{BE} \text{-----(1)}$$

Solve for V_C

$$V_C = -V_{BE} \left(gm + \frac{1}{R_{eq}} + sC_{in} + sC_6 \right) \text{-----(2)}$$

Node 2:

$$\frac{V_C}{C} + (V_C + V_{BE})(sC_6) + gm(V_{BE}) = 0 \text{---(3)}$$

$$\frac{-V_C}{Z_{eq}} - V_C(sC_6) - V_{BE}(sC_6) - gm(V_{BE}) = 0 \text{---(4)}$$

Substitute V_C with equation (2).

$$\frac{V_{BE}(gm + \frac{1}{R_{eq}} + sC_{in} + sC_6)}{Z_{eq}} + V_{BE} \left(gm + \frac{1}{R_{eq}} + sC_{in} + sC_6 \right) (sC_6) - V_{BE}(sC_6) - gm(V_{BE}) = 0 \text{---(5)}$$

Separating the imaginary part of equation (5) to solve for ω gives:

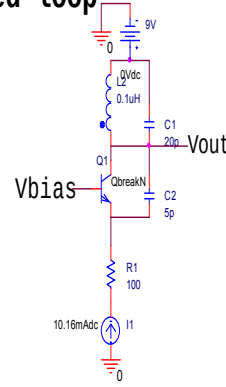
$$j \left(\frac{-gm}{L} + \omega^2 gm C_{eq} - \frac{1}{LR_{eq}} + \frac{\omega^2 C_{eq}}{R_{eq}} + \omega^2 gm C_6 + \frac{\omega^2 C_6}{R_{eq}} - \omega^2 C_6^2 \right) = 0$$

$$\omega = \sqrt{\frac{gm R_{eq} + 1}{LR_{eq} [C_{eq} (gm + \frac{1}{R_{eq}}) + C_6 (gm + \frac{1}{R_{eq}} - C_6)]}}$$

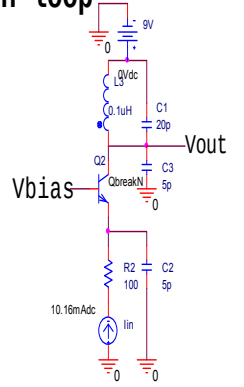
$$f = \frac{1}{2\pi} \sqrt{\frac{gm R_{eq} + 1}{LR_{eq} [C_{eq} (gm + \frac{1}{R_{eq}}) + C_6 (gm + \frac{1}{R_{eq}} - C_6)]}} \approx \frac{1}{2\pi \sqrt{LC_5}}$$

Trans-Impedance Calculations (Stage 2):

Closed loop



Open loop



$$\beta = \frac{1}{Z_c}, \quad R_{in\ CB} = \frac{1}{g_m}, \quad g_m = \frac{I_C}{V_E} = \frac{10.16mA}{26mV} = .3908$$

$$\frac{V_o}{i_{in}} = R = A_{open\ loop}$$

$$I_x = i_c \left(\frac{Z_c + Z_R}{Z_c} \right)$$

$$I_x = 10.16mA \left(\frac{\frac{1}{C_s} + 100}{\frac{1}{C_s}} \right) = 10.16mA [1 + C_s(100)]$$

$$R_{in} = \frac{1}{g_m} \parallel R_E = \frac{\frac{R_E}{g_m}}{R_E + \frac{1}{g_m}} = \frac{R_E}{g_m R_E + 1}$$

$$R_E = \frac{\frac{100}{C_s}}{\frac{1}{C_s} + 100} = \frac{100}{1 + C_s 100}$$

$$\begin{aligned} \frac{V_o}{i_s} &= \frac{i_x R_{in\ CB}}{i_{in}} = \frac{i_{in} \frac{Z_c + Z_R}{Z_c}}{i_{in}} * \left(\frac{R_E}{R_E g_m + 1} \right)_{CB} \\ &= \frac{Z_c + Z_R}{Z_c} * \frac{\frac{Z_c * Z_R}{Z_c + Z_R}}{g_m \left(\frac{Z_c * Z_R}{Z_c + Z_R} \right)} \end{aligned}$$

$$\frac{\frac{Z_R}{g_m Z_C Z_R + Z_C Z_R}}{Z_R + Z_C} = \frac{Z_C + Z_R}{Z_C (g_m + 1)} = R_{in}' = A_{RO}$$

$$= A_{r \text{ open loop}}$$

$$A_{RO} = \frac{V_o R_{in}'}{V_{in}}$$

$$\frac{V_o}{V_n} = A_{V \text{ CB}} = Z_{load} g_m = Z_{eq} g_m$$

$$\frac{V_o}{V_{in}} * \frac{V_{in}}{i_{in}} = \left[g_m \frac{Z_{(C_1+C_2)} Z_L}{Z_{(C_1+C_2)} + Z_L} \right] * \left[\frac{Z_C + Z_R}{Z_C (g_m + 1)} \right]$$

$$= A_{RO}$$

$$A_{RC} = A_{RO} \left(\frac{A_o}{1 + \beta A_o} \right) = \frac{A_o}{1 + \frac{1}{Z_{C_1}} A_o}$$